

All-Genus 2D $\mathcal{N} = 1$ Superconformal Blocks

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Abstract

We develop a numerical algorithm for two-dimensional $\mathcal{N} = 1$ superconformal blocks at arbitrary genera, in arbitrary plumbing channels, and for arbitrary spin structures. The algorithm utilizes two commuting copies of Virasoro subalgebras of the direct sum of the $\mathcal{N} = 1$ superconformal algebra with an auxiliary free fermion, which allows us to reduce the $\mathcal{N} = 1$ superconformal blocks to Virasoro conformal blocks.

1 Introduction

Superconformal blocks are the central objects for computing correlators of superconformal field theories (SCFTs). Given the superconformal block on a spin Riemann surface, together with the conformal data of the SCFT, namely the conformal weights and the structure constants, correlators of a 2-dimensional $\mathcal{N} = 1$ SCFT on that spin Riemann surface can be computed through the superconformal block decomposition.

On spin Riemann surfaces of genus $g = 0$ or $g = 1$, algorithms for numerically computing the superconformal blocks are well developed. Analogously to the Zamolodchikov recursion relation for the sphere 4-point Virasoro conformal block [1, 2], the sphere 4-point superconformal block on a sphere can be computed using recursion relations in the central charge c or in the intermediate conformal weights h . These recursion relations have been worked out for both the case with all-NS external states [3–5] and that with Ramond external states [6, 7]. The c -recursion relation can be generalized to arbitrary plumbing channels and arbitrary genus as long as all intermediate states are in the NS sector [8, 9]. Meanwhile, for the blocks that involve Ramond intermediate states, the h -recursion relations can only be generalized to genus 1 [10] while the c -recursion relation does not exist beyond tree level, due to the fact that the large- h or large- c limit of the Ramond conformal blocks does not exist at higher genus. Algorithms for numerically computing these higher-genus Ramond blocks are absent in the literature.

In this work, we present the first numerical algorithm for these higher-genus Ramond blocks. The algorithm is based on the fact that the $\mathcal{N} = 1$ superconformal algebra $\mathbf{SCA}$, direct summed with an auxiliary free fermion algebra $\mathbf{F}$, admits two copies of commuting Virasoro subalgebras [11, 12]. These subalgebras were first discovered in the context of the AGT relation [13, 14] and have been used in the context of the super-Liouville/double Liouville correspondence [15, 16]. Using this $\mathbf{Vir} \oplus \mathbf{Vir}$ subalgebra, we can reduce the superconformal block to a sum of products of two Virasoro conformal blocks, which can be computed numerically as a series in the plumbing parameters using the c -recursion relation developed in [8].

The rest of this paper is organized as follows. In Section 2, we define the superconformal blocks using the plumbing fixture. In Section 3, we review the details of the $\mathbf{Vir} \oplus \mathbf{Vir}$ subalgebra and the branching rules of $\mathbf{SCA} \oplus \mathbf{F}$ supermultiplets. In Section 4, we derive the relation between superconformal blocks and Virasoro blocks for blocks without a certain type of obstruction which we call odd Ramond loops. Section 5 extends the construction to blocks with odd Ramond loops using an auxiliary fermion insertion. We present numerical tests for the algorithm in Section 6 and discuss applications and future directions in Section 7.

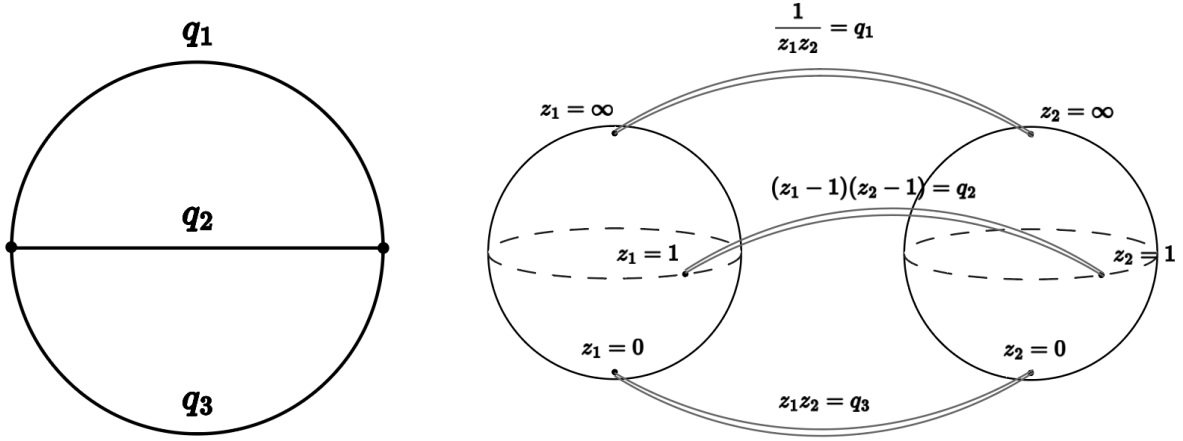


Figure 1: The genus-2 "theta" channel, where the left picture shows the trivalent graph Γ , and the right picture shows the plumbing performed to construct a genus-2 Riemann surface.

2 The superconformal blocks

An arbitrary punctured Riemann surface can be constructed by gluing 3-punctured Riemann spheres using the plumbing fixture, which takes two punctures with local coordinates z_1, z_2 and identifies a pair of annular regions using

$$z_1 z_2 = q, \quad |q| \leq 1, \quad (1)$$

where q is a complex plumbing parameter. A genus- g , n -punctured Riemann surface can be constructed by plumbing $(3g - 3 + n)$ pairs of punctures, and its $(3g - 3 + n)$ complex moduli are given by those $(3g - 3 + n)$ plumbing parameters. One can then describe a punctured Riemann surface using a trivalent graph Γ together with a q -parameter for each internal edge. The vertices of Γ correspond to the 3-punctured spheres, while the edges connect pairs of plumbed punctures. We call this graph Γ a plumbing channel. An example of such a construction, the genus-2 theta channel, is shown in figure 1.

To specify a spin structure on the Riemann surface, we assign a sign η_e for each internal edge e in Γ , and additionally label each edge of Γ by NS or R. In the $\mathcal{N} = 1$ SCFT language, η_e specifies whether there is a line operator insertion $(-1)^F$ around the tube associated with edge e , while the NS/R label specifies the fermionic boundary conditions of the states propagating along the tube. At each vertex, only edges of types (NS, NS, NS) or (NS, R, R) can be connected together. We denote the plumbing graph with the NS/R labels by $\hat{\Gamma}$, and the data of the spin Riemann surface is specified by $(\hat{\Gamma}, \{q_e, \eta_e\})$. The sets of all NS internal edges, R internal edges, (NS, NS, NS) vertices and (NS, R, R) vertices in $\hat{\Gamma}$ are denoted by $E_{\text{NS}}, E_{\text{R}}, V_{\text{NS}}$ and V_{R} , respectively. We also include in $\hat{\Gamma}$ the order of the three half-edges that connect to each

vertex, with the first half-edge corresponding to the operator inserted at $z = \infty$, the second corresponding to $z = 1$, and the third corresponding to $z = 0$. On a (NS, R, R) sphere, we always take the NS half-edge to be in the first slot.

Plumbing two punctures in an SCFT correlator amounts to summing over a complete basis of intermediate states. These states organize into supermultiplets. We denote an NS superconformal primary by ϕ . A Ramond supermultiplet has 2 ground states $w^\pm$ that are related by G_0 :

$$G_0 w^\pm = i\beta e^{\mp i\pi/4} w^\mp, \quad (2)$$

where β is defined by $\beta^2 = \frac{c}{24} - h$. We take w^+ to be Grassmann even, and hence w^- will be Grassmann odd. A basis of states in a supermultiplet is given by

$$\begin{aligned} \mathbf{L}_{-A}\phi &\equiv L_{-n_1} \cdots L_{-n_\ell} G_{-r_1} \cdots G_{-r_p} \phi, \\ \mathbb{L}_{-A}w^\pm &\equiv L_{-n_1} \cdots L_{-n_\ell} G_{-r_1} \cdots G_{-r_p} w^\pm, \end{aligned} \quad (3)$$

in the NS and R sectors, respectively. Here, $n_i \in \mathbb{Z}$, $r_j \in \mathbb{Z} + \nu$, and $n_1 \geq \dots \geq n_\ell > 0$, $r_1 > \dots > r_p > 0$, where $\nu = 0$ in the R sector and $\nu = \frac{1}{2}$ in the NS sector. We further define the total level of $\mathbf{L}_{-A}, \mathbb{L}_{-A}$ by $|A| \equiv \sum_{i=1}^\ell n_i + \sum_{j=1}^p r_j$, and their Grassmann parity by $(-1)^A \equiv (-1)^p$.

After replacing each pair of plumbed punctures by a pair of local operators, we can reduce the SCFT correlator to a product of 3-point functions on the spheres. On an (NS, NS, NS) sphere, all these 3-point functions of descendants can be reduced to either $\langle \phi_1 \phi_2 \phi_3 \rangle$ or $\langle \phi_1 (G_{-\frac{1}{2}} \phi_2) \phi_3 \rangle$ by superconformal Ward identities. We define the normalized 3-point functions ρ_a by

$$\rho_0(\phi_1, \phi_2, \phi_3) = 1, \quad \rho_1(\phi_1, G_{-\frac{1}{2}} \phi_2, \phi_3) = 1, \quad (4)$$

where the operators ϕ_1, ϕ_2, ϕ_3 are inserted at $\infty, 1, 0$, respectively, and the operator at the first slot is inserted using the inversion map $w = 1/z$. The normalized three-point functions $\rho_a(\mathbf{L}_{-A}\phi_1, \mathbf{L}_{-B}\phi_2, \mathbf{L}_{-C}\phi_3)$ of descendants are defined by reducing them to $\rho_0(\phi_1, \phi_2, \phi_3)$, $\rho_1(\phi_1, G_{-\frac{1}{2}} \phi_2, \phi_3)$ using superconformal Ward identities. On an (NS, R, R) sphere, there are 4 independent 3-point functions, and we define

$$\begin{aligned} \rho_0^{(\eta)}(\phi_1, w_2^+, w_3^+) &= 1, & \rho_0^{(\eta)}(\phi_1, w_2^-, w_3^-) &= \eta, \\ \rho_1^{(\eta)}(\phi_1, w_2^+, w_3^-) &= 1, & \rho_1^{(\eta)}(\phi_1, w_2^-, w_3^+) &= i\eta, \end{aligned} \quad (5)$$

for $\eta = \pm$. Again, the operators $\phi_1, w_2^\pm, w_3^\pm$ are inserted at $\infty, 1, 0$, respectively, and the normalized 3-point functions of descendants are defined through superconformal Ward identities.

On a spin Riemann surface specified by the plumbing data $(\hat{\Gamma}, \{q_e, \eta_e\})$, the superconformal

block with internal weights $\{h_e\}_{e \in E_{\text{NS}}}, \{\beta_e\}_{e \in E_{\text{R}}}$ is defined as

$$\begin{aligned}
\mathbf{F}_{\hat{\Gamma}}^{\{f, \eta, a\}}(h_e, \beta_e; q_e, \eta_e) &= \sum_{\{A, \alpha\}} (-1)^{K_{\Gamma}(\epsilon)} \prod_{e \in E_{\text{NS}} \cup E_{\text{R}}} q_e^{|A_{e_1}|} \eta_e^{\epsilon_e} \\
&\times \prod_{e \in E_{\text{NS}}} \mathbf{G}_{h_e}^{A_{e_1} A_{e_2}} \prod_{e \in E_{\text{R}}} \mathbb{G}_{\beta_e}^{A_{e_1} \alpha_{e_1}, A_{e_2} \alpha_{e_2}} \\
&\times \prod_{v \in V_{\text{NS}}} \rho_{a_v}(\mathbf{L}_{-A_{v_1}} \phi_{v_1}, \mathbf{L}_{-A_{v_2}} \phi_{v_2}, \mathbf{L}_{-A_{v_3}} \phi_{v_3}) \\
&\times \prod_{v \in V_{\text{R}}} \rho_{f_v}^{(\eta_v)}(\mathbf{L}_{-A_{v_1}} \phi_{v_1}, \mathbb{L}_{-A_{v_2}} w_{v_2}^{\alpha_{v_2}}, \mathbb{L}_{-A_{v_3}} w_{v_3}^{\alpha_{v_3}}),
\end{aligned} \tag{6}$$

where f_v, η_v label the 3-point functions on the (NS, R, R) spheres, and a_v labels those on the (NS, NS, NS) spheres. The two half-edges an edge e contains are labeled e_1, e_2 , and the three half-edges connected to a vertex v are labeled v_1, v_2, v_3 according to their aforementioned order ¹. The sum is over all displayed descendant labels A and ground state labels α on each half-edge, subject to the conditions

$$\begin{aligned}
|A_{e_1}| &= |A_{e_2}|, \quad e \in E_{\text{NS}} \cup E_{\text{R}} \\
(-1)^{A_{v_1} + A_{v_2} + A_{v_3}} &= (-1)^{a_v}, \quad v \in V_{\text{NS}}, \\
(-1)^{A_{v_1} + A_{v_2} + A_{v_3} + \alpha_{v_2} + \alpha_{v_3}} &= (-1)^{f_v}, \quad v \in V_{\text{R}}.
\end{aligned} \tag{7}$$

We define ϵ_{v_i} to be the total Grassmann parity of the descendant Φ_{v_i} , which stands for $\mathbf{L}_{-A_{v,i}} \phi_{v_i}$ in the NS sector or $\mathbb{L}_{-A_{v,i}} w_{v_i}^{\alpha_{v_i}}$ in the R sector. The sign $(-1)^{K_{\Gamma}(\epsilon)}$ is the Grassmann sign that arises from moving the operator order from the pairwise insertion order to the displayed order in the 3-point functions ². $\mathbf{G}, \mathbb{G}$ are inverse Gram matrices in the NS and R sectors, which are defined as the inverses of

$$\begin{aligned}
(\mathbf{G}_h)_{AB} &= \langle \langle \mathbf{L}_{-A} \phi_h | \mathbf{L}_{-B} \phi_h \rangle \rangle, \\
(\mathbb{G}_{\beta})_{A\alpha, B\alpha'} &= \langle \langle \mathbb{L}_{-A} w_{\beta}^{\alpha} | \mathbb{L}_{-B} w_{\beta}^{\alpha'} \rangle \rangle,
\end{aligned} \tag{9}$$

where the double left bracket stands for BPZ conjugation, and the conventions we use for which are summarized in Appendix A. h, β label the conformal weight and G_0 eigenvalues of the primaries, which are normalized such that $\langle \langle \phi_h | \phi_h \rangle \rangle = 1$, $\langle \langle w_{\beta}^+ | w_{\beta}^+ \rangle \rangle = 1$. In the rest of this paper, we will omit the labels $(h_e, \beta_e; q_e, \eta_e)$ when writing the superconformal block.

¹The order of the two half-edges in an edge does not matter, since the Grassmann sign of exchanging Φ_A, Φ_B is cancelled by the relative sign between $\mathbf{G}^{AB}$ and $\mathbf{G}^{BA}$.

²More explicitly, this is the Grassmann sign that arises from moving the operators from order $\prod_{e \in E_{\text{NS}} \cup E_{\text{R}}} \Phi_{e_1} \Phi_{e_2}$ to order $\prod_{v \in V_{\text{NS}}} \Phi_{v_1} \Phi_{v_2} \Phi_{v_3} \prod_{v \in V_{\text{R}}} \Phi_{v_1} \Phi_{v_2} \Phi_{v_3}$. It can be defined as

$$K_{\Gamma}(\epsilon) = \sum_{(v_i, w_j)} \epsilon_{v_i} \epsilon_{w_j} \pmod{2}, \tag{8}$$

where the sum is over pairs (v_i, w_j) such that v_i precedes w_j in the pairwise insertion but follows w_j in the order of the 3-point functions.

3 The double Virasoro subalgebra

The key fact we utilize is that the enlarged algebra $\text{SCA}_\nu \oplus \text{F}_\nu$ constructed from direct summing the $\mathcal{N} = 1$ superconformal algebra SCA_ν with shift $\nu \in \{0, \frac{1}{2}\}$ with an auxiliary free fermion algebra F_ν with the same shift ν

$$\{\psi_r, \psi_s\} = \delta_{r+s,0}, \quad \{\psi_r, G_s\} = [\psi_r, L_n] = 0, \quad (10)$$

admits two commuting Virasoro subalgebras [11, 12, 16].

The generators of the two commuting Virasoro algebras can be written down explicitly as the following. We introduce the following operators

$$L_n^{\text{F}} = \frac{1}{2} \sum_{r \in \mathbb{Z} + \nu} r : \psi_{n-r} \psi_r : + \frac{1-2\nu}{16} \delta_{n,0}, \quad U_n = \sum_{r \in \mathbb{Z} + \nu} \psi_{n-r} G_r, \quad (11)$$

and the Virasoro generators are given by

$$L_n^{(1)} = \frac{b^{-1} L_n - (b^{-1} + 2b) L_n^{\text{F}} + U_n}{b^{-1} - b}, \quad L_n^{(2)} = \frac{b L_n - (b + 2b^{-1}) L_n^{\text{F}} + U_n}{b - b^{-1}}, \quad (12)$$

where b is defined by the SCA central charge $c = \frac{3}{2} + 3(b + b^{-1})^2$, and we have assumed that $b \neq 0, 1$. These generators $L_n^{(1,2)}$ satisfy $L_n^{(1)} + L_n^{(2)} = L_n + L_n^{\text{F}}$, and the two central charges of the two Virasoro algebras are given by

$$c^{(1)} = 1 + \frac{3Q^2}{1 - b^2}, \quad c^{(2)} = 1 + \frac{3Q^2}{1 - b^{-2}}, \quad (13)$$

where the Liouville background charge Q is defined such that $Q = b + b^{-1}$.

Let $\mathcal{M}_{\text{NS}}(h), \mathcal{M}_{\text{R}}(\beta)$ denote the tensor products of the auxiliary fermion Fock space with the corresponding SCFT Verma module. We denote the two R-sector ground states for the free fermion by u^a with $a = 0, 1$, which satisfy

$$\psi_0 u^a = \frac{1}{\sqrt{2}} u^{1-a}. \quad (14)$$

The highest weight states in $\mathcal{M}_{\text{NS}}(h), \mathcal{M}_{\text{R}}(\beta)$ are respectively $\mathbf{1} \otimes \phi$ and $u^a \otimes w^a$. For Liouville momentum $P \notin \frac{1}{2}(b\mathbb{Z} + b^{-1}\mathbb{Z})$, these representations have branching rules [11, 12]

$$\begin{aligned} \mathcal{M}_{\text{NS}}(h) &\cong \bigoplus_{n \in \frac{1}{2}\mathbb{Z}} V_{c^{(1)}}(h_n^{(1)}) \otimes V_{c^{(2)}}(h_n^{(2)}), \\ \mathcal{M}_{\text{R}}(\beta) &\cong \bigoplus_{\alpha=0,1} \bigoplus_{n \in \frac{1}{2}\mathbb{Z} + \frac{1}{4}} V_{c^{(1)}}(h_n^{(1)}) \otimes V_{c^{(2)}}(h_n^{(2)}), \end{aligned} \quad (15)$$

where P is defined using $P^2 = \frac{1}{4}Q^2 - 2h$ in the NS sector and $P^2 = 2\beta^2$ in the R sector. Here, $V_c(h)$ is the Virasoro Verma module with highest weight h and central charge c . The highest-weight vector in $V_{c(1)}(h_n^{(1)}) \otimes V_{c(2)}(h_n^{(2)})$ is denoted by v_n and v_n^α in NS and R sectors, respectively. Their conformal weights are

$$h_n^{(1)} = \frac{Q^2/4 - (P + 2nb)^2}{2(1 - b^2)}, \quad h_n^{(2)} = \frac{Q^2/4 - (P + 2nb^{-1})^2}{2(1 - b^{-2})}. \quad (16)$$

The NS $\text{Vir} \oplus \text{Vir}$ primary v_n has Grassmann parity $2n \bmod 2$ and level $2n^2$ relative to $\mathbf{1} \otimes \phi$, and the R primary v_n^α has Grassmann parity $\alpha = 0, 1$ and level $2n^2 - \frac{1}{8}$ relative to $u^0 \otimes w^0$. These $\text{Vir} \oplus \text{Vir}$ primaries can be realized by a free field representation, which we review in appendix B.

4 Reduction of superconformal blocks

To reduce the superconformal block (6) to Virasoro conformal blocks, we first consider an enlarged $\text{SCA} \oplus \text{F}$ block $\widehat{\mathbf{F}}_{\hat{\mathbf{F}}}^{\{f, \eta, a\}}$, defined as

$$\begin{aligned} \widehat{\mathbf{F}}_{\hat{\mathbf{F}}}^{\{f, \eta, a\}} &= \sum_{\{A, \alpha, \mathbf{A}, \mathbf{a}\}} (-1)^{K_{\Gamma}(\epsilon + \epsilon')} \prod_{e \in E_{\text{NS}} \cup E_{\text{R}}} q_e^{|A_{e_1}| + |A_{e_1}|} \eta_e^{\epsilon_e + \epsilon'_e} (-1)^{\epsilon_e \epsilon'_e} \\ &\times \prod_{e \in E_{\text{NS}}} \mathbf{G}_{\mathbf{F}}^{A_{e_1} A_{e_2}} \mathbf{G}_{h_e}^{A_{e_1} A_{e_2}} \prod_{e \in E_{\text{R}}} \mathbb{G}_{\mathbf{F}}^{A_{e_1} \mathbf{a}_{e_1}, A_{e_2} \mathbf{a}_{e_2}} \mathbb{G}_{\beta_e}^{A_{e_1} \alpha_{e_1}, A_{e_2} \alpha_{e_2}} \\ &\times \prod_{v \in V_{\text{NS}}} \hat{\rho}_{a_v}(\Psi_{-\mathbf{A}_{v_1}} \mathbf{1} \otimes \mathbf{L}_{-A_{v_1}} \phi_{v_1}, \Psi_{-\mathbf{A}_{v_2}} \mathbf{1} \otimes \mathbf{L}_{-A_{v_2}} \phi_{v_2}, \Psi_{-\mathbf{A}_{v_3}} \mathbf{1} \otimes \mathbf{L}_{-A_{v_3}} \phi_{v_3}) \\ &\times \prod_{v \in V_{\text{R}}} \hat{\rho}_{f_v}^{(\eta_v)}(\Psi_{-\mathbf{A}_{v_1}} \mathbf{1} \otimes \mathbf{L}_{-A_{v_1}} \phi_{v_1}, \Psi_{-\mathbf{A}_{v_2}} u^{\mathbf{a}_{v_2}} \otimes \mathbb{L}_{-A_{v_2}} w_{v_2}^{\alpha_{v_2}}, \Psi_{-\mathbf{A}_{v_3}} u^{\mathbf{a}_{v_3}} \otimes \mathbb{L}_{-A_{v_3}} w_{v_3}^{\alpha_{v_3}}) \end{aligned} \quad (17)$$

where $\Psi_{-\mathbf{A}}$ denotes a chain of fermion creation operators ψ_{-r} with $r > 0$, and $|\mathbf{A}|$ denotes its total level. The sum is over all displayed descendant and ground state labels, subject to

$$\begin{aligned} A_{e_1} &= A_{e_2}, \quad e \in E_{\text{NS}} \cup E_{\text{R}} \\ \mathbf{a}_{e_1} &= \mathbf{a}_{e_2}, \quad e \in E_{\text{R}} \\ (-1)^{A_{v_1} + A_{v_2} + A_{v_3}} &= 1, \quad v \in V_{\text{NS}} \\ (-1)^{A_{v_1} + A_{v_2} + A_{v_3} + \mathbf{a}_{v_2} + \mathbf{a}_{v_3}} &= 1, \quad v \in V_{\text{R}}, \end{aligned} \quad (18)$$

in addition to (7). ϵ and ϵ' denote the Grassmann parity of the SCA and the fermion part of the corresponding descendant, respectively. The fermion Gram matrices are given by

$$\begin{aligned} (\mathbf{G}_{\mathbf{F}})_{\mathbf{A}\mathbf{B}} &= \langle \langle \Psi_{-\mathbf{A}} \mathbf{1} | \Psi_{-\mathbf{B}} \mathbf{1} \rangle \rangle, \\ (\mathbb{G}_{\mathbf{F}})_{\mathbf{A}\mathbf{a}, \mathbf{B}\mathbf{b}} &= \langle \langle \Psi_{-\mathbf{A}} u^{\mathbf{a}} | \Psi_{-\mathbf{B}} u^{\mathbf{b}} \rangle \rangle. \end{aligned} \quad (19)$$

The enlarged 3-point functions $\hat{\rho}_a, \hat{\rho}_f^{(\eta)}$ are defined using the product of the fermion 3-point function and the normalized 3-point functions as

$$\begin{aligned} & \hat{\rho}_a(\Psi_{-A}\mathbf{1} \otimes \mathbf{L}_{-A}\phi_1, \Psi_{-B}\mathbf{1} \otimes \mathbf{L}_{-B}\phi_2, \Psi_{-C}\mathbf{1} \otimes \mathbf{L}_{-C}\phi_3) \\ &= (-1)^{\epsilon_1\epsilon'_1+\epsilon_2\epsilon'_2}\rho_F(\Psi_{-A}\mathbf{1}, \Psi_{-B}\mathbf{1}, \Psi_{-C}\mathbf{1})\rho_a(\mathbf{L}_{-A}\phi_1, \mathbf{L}_{-B}\phi_2, \mathbf{L}_{-C}\phi_3), \\ & \hat{\rho}_f^{(\eta)}(\Psi_{-A}\mathbf{1} \otimes \mathbf{L}_{-A}\phi_1, \Psi_{-B}u^b \otimes \mathbb{L}_{-B}w_2^\alpha, \Psi_{-C}u^c \otimes \mathbb{L}_{-C}w_3^\gamma) \\ &= (-1)^{\epsilon_1\epsilon'_1+\epsilon_2\epsilon'_2}\rho_F(\Psi_{-A}\mathbf{1}, \Psi_{-B}u^b, \Psi_{-C}u^c)\rho_f^{(\eta)}(\mathbf{L}_{-A}\phi_1, \mathbb{L}_{-B}w_2^\alpha, \mathbb{L}_{-C}w_3^\gamma), \end{aligned} \quad (20)$$

where the fermion 3-point function is normalized by $\rho_F(\mathbf{1}, \mathbf{1}, \mathbf{1}) = 1$, $\rho_F(\mathbf{1}, u^0, u^0) = 1$ and $\rho_F(\mathbf{1}, u^1, u^1) = i$.

This enlarged block $\hat{\mathbf{F}}_{\hat{\Gamma}}^{\{f, \eta, a\}}$ can be computed in two different ways. One can take the basis $\Psi_{-A}\mathbf{1} \otimes \mathbf{L}_{-A}\phi$ for states in $\mathcal{M}_{\text{NS}}(h)$ and $\Psi_{-A}u^a \otimes \mathbb{L}_{-A}w^\alpha$ for $\mathcal{M}_{\text{R}}(\beta)$. This allows us to express the enlarged block in terms of the superconformal block (6) and the free fermion block. Namely,

$$\hat{\mathbf{F}}_{\hat{\Gamma}}^{\{f, \eta, a\}} = \mathbf{F}_{\hat{\Gamma}}^{\{f, \eta, a\}} \star_{\hat{\Gamma}} \mathbf{F}_{\hat{\Gamma}}^{\text{F}}, \quad (21)$$

where we view the blocks as formal series in q_e, η_e , and the convolution $\star_{\hat{\Gamma}}$ acts on monomials as

$$\begin{aligned} & \left(\prod_e q_e^{r_e} \eta_e^{\epsilon_e} \right) \star_{\hat{\Gamma}} \left(\prod_e q_e^{s_e} \eta_e^{\epsilon'_e} \right) \\ &= (-1)^{K_{\Gamma}(\epsilon+\epsilon')-K_{\Gamma}(\epsilon)-K_{\Gamma}(\epsilon')+\sum_e \epsilon_e \epsilon'_e + \sum_v (\epsilon_{v,1} \epsilon'_{v,1} + \epsilon_{v,2} \epsilon'_{v,3})} \prod_e q_e^{r_e+s_e} \eta_e^{\epsilon_e+\epsilon'_e}, \end{aligned} \quad (22)$$

while the free fermion block $\mathbf{F}_{\text{F}}$ is defined as

$$\begin{aligned} \mathbf{F}_{\hat{\Gamma}}^{\text{F}} &= \sum_{\{A, \mathbf{a}\}} (-1)^{K_{\Gamma}(\epsilon')} \prod_{e \in E_{\text{NS}} \cup E_{\text{R}}} q_e^{|\mathbf{A}_{e1}|} \eta_e^{\epsilon'_e} \prod_{e \in E_{\text{NS}}} \mathbf{G}_{\text{F}}^{\mathbf{A}_{e1} \mathbf{A}_{e2}} \prod_{e \in E_{\text{R}}} \mathbb{G}_{\beta_e}^{\mathbf{A}_{e1} \mathbf{a}_{e1}, \mathbf{A}_{e2} \mathbf{a}_{e2}} \\ &\times \prod_{v \in V_{\text{NS}}} \rho_F(\Psi_{-A_{v1}}\mathbf{1}, \Psi_{-A_{v2}}\mathbf{1}, \Psi_{-A_{v3}}\mathbf{1}) \prod_{v \in V_{\text{R}}} \rho_F(\Psi_{-A_{v1}}\mathbf{1}, \Psi_{-A_{v2}}u^{\mathbf{a}_{v2}}, \Psi_{-A_{v3}}u^{\mathbf{a}_{v3}}). \end{aligned} \quad (23)$$

As is, the enlarged block $\hat{\mathbf{F}}_f^{(\eta, \eta')}$ vanishes whenever the vertex signs η_v around a closed Ramond loop $C \subset \hat{\Gamma}$ have product

$$\prod_{v \in C} (-1)^{f_v} \eta_v = -1. \quad (24)$$

We call such a loop an odd Ramond loop, and the algorithm for computing superconformal blocks with odd Ramond loops will be presented in Section 5. In the rest of this section, we assume that there are no such odd Ramond loops.

The enlarged block can also be computed using the basis $L_{-A}^{(1)}L_{-B}^{(2)}v_n$ for $\mathcal{M}_{\text{NS}}(h)$ and $L_{-A}^{(1)}L_{-B}^{(2)}v_n^\alpha$ for $\mathcal{M}_{\text{R}}(\beta)$. This leads to the following decomposition of $\widehat{\mathbf{F}}_{\hat{\Gamma}}^{\{f,\eta,a\}}$:

$$\begin{aligned}\widehat{\mathbf{F}}_{\hat{\Gamma}}^{\{f,\eta,a\}} &= \sum_{\{n,\alpha\}} (-1)^{K_{\Gamma}(\epsilon)} \prod_{e \in E_{\text{NS}}} q_e^{2n_e^2} \eta_e^{2n_e} \prod_{e \in E_{\text{R}}} q_e^{2n_e^2 - \frac{1}{8}} \eta_e^{\alpha_e} \\ &\times \prod_{i=1}^2 F_{\Gamma}(h_{e,n_e}^{(i)}, c^{(i)}) \prod_{v \in V_{\text{NS}}} \mathbf{B}_a(n_{v_1}, n_{v_2}, n_{v_3}) \\ &\times \prod_{v \in V_{\text{R}}} \mathbb{B}_f^{(\eta_v)}(n_{v_1}, n_{v_2}, n_{v_3}, \alpha_{v_2}, \alpha_{v_3}),\end{aligned}\tag{25}$$

where the labels n take values in $n \in \frac{1}{2}\mathbb{Z}$ for NS half-edges, and $n \in \frac{1}{2}\mathbb{Z} + \frac{1}{4}$ for R half-edges. $F(h_e; c)$ denotes the Virasoro conformal block in the same channel Γ with intermediate weights h_e and central charge c . ϵ here denotes the Grassmann parity of the $\mathbf{Vir} \oplus \mathbf{Vir}$ primary v_n or v_n^α . $\mathbf{B}_a, \mathbb{B}_f^{(\eta)}$ are the branching coefficients, defined as

$$\begin{aligned}\mathbf{B}_a(n_1, n_2, n_3) &= \frac{\hat{\rho}_a(v_{1,n_1}, v_{2,n_2}, v_{3,n_3})}{\|v_{1,n_1}\| \|v_{2,n_2}\| \|v_{3,n_3}\|}, \\ \mathbb{B}_f^{(\eta)}(n_1, n_2, n_3, \alpha_2, \alpha_3) &= \frac{\hat{\rho}_f^{(\eta)}(v_{1,n_1}, v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3})}{\|v_{1,n_1}\| \|v_{2,n_2}^{\alpha_2}\| \|v_{3,n_3}^{\alpha_3}\|}.\end{aligned}\tag{26}$$

The (NS, NS, NS) branching coefficients have been worked out in [16], while the (NS, R, R) coefficients have never been computed analytically in the literature. We now present a recursive algorithm for determining both numerically. It can be proved that, for $n > 0$, $L_1 v_n, L_1 v_n^\alpha$ both only contain level- $(4n-3)$ states in the $\mathbf{Vir} \oplus \mathbf{Vir}$ module generated by v_{n-1}, v_{n-1}^α , while $L_{-1} v_n, L_{-1} v_n^\alpha$ only contain level-1 states in the v_n, v_n^α module as well as level- $(4n-1)$ states in the v_{n-1}, v_{n-1}^α module. Namely, there exist coefficients $\mathbf{V}_n^{(1,2)}, \mathbf{V}_{\pm 1,n}^{AB}, \mathbb{V}_n^{(1,2)}, \mathbb{V}_{\pm 1,n}^{AB}$ such that

$$\begin{aligned}L_1 v_n^\alpha &= \sum_{|A|+|B|=4n-3} \mathbb{V}_{1,n,\alpha}^{AB} L_{-A}^{(1)} L_{-B}^{(2)} v_{n-1}^\alpha, \\ L_{-1} v_n^\alpha &= \mathbb{V}_{n,\alpha}^{(1)} L_{-1}^{(1)} v_n^\alpha + \mathbb{V}_{n,\alpha}^{(2)} L_{-1}^{(2)} v_n^\alpha + \sum_{|A|+|B|=4n-1} \mathbb{V}_{-1,n,\alpha}^{AB} L_{-A}^{(1)} L_{-B}^{(2)} v_{n-1}^\alpha, \\ L_1 v_n &= \sum_{|A|+|B|=4n-3} \mathbf{V}_{1,n}^{AB} L_{-A}^{(1)} L_{-B}^{(2)} v_{n-1}, \\ L_{-1} v_n &= \mathbf{V}_n^{(1)} L_{-1}^{(1)} v_n + \mathbf{V}_n^{(2)} L_{-1}^{(2)} v_n + \sum_{|A|+|B|=4n-1} \mathbf{V}_{-1,n}^{AB} L_{-A}^{(1)} L_{-B}^{(2)} v_{n-1}.\end{aligned}\tag{27}$$

These coefficients $\mathbf{V}_n^{(1,2)}, \mathbf{V}_{\pm 1,n}^{AB}, \mathbb{V}_n^{(1,2)}, \mathbb{V}_{\pm 1,n}^{AB}$ can be determined directly by applying the free-field realization of $L_{\pm 1}$ on (53), (55). A similar relation exists for $n < 0$, where $L_{\pm 1} v_n, L_{\pm 1} v_n^\alpha$ include states in the v_{n+1}, v_{n+1}^α module instead of v_{n-1}, v_{n-1}^α .

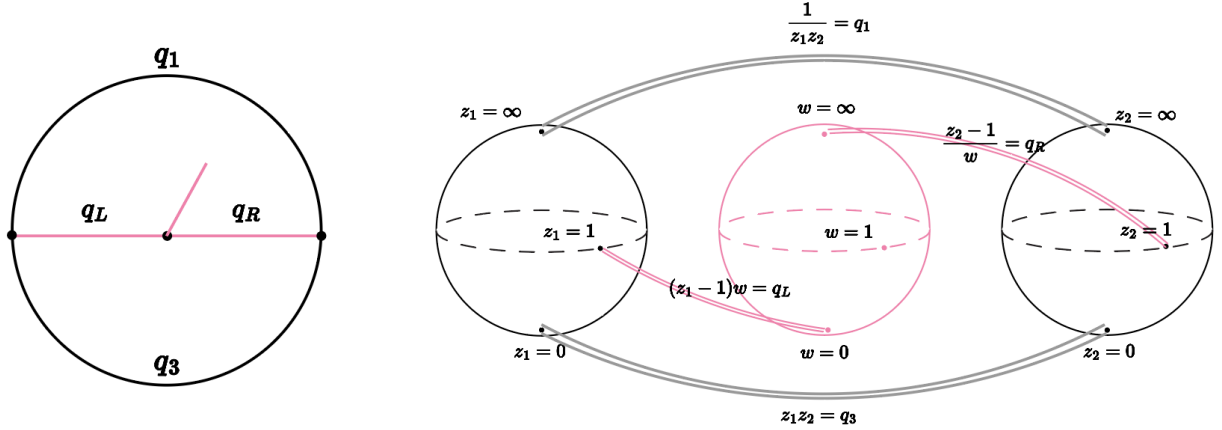


Figure 2: Splitting the second edge in the genus-2 theta channel. The second edge that connects $z_1 = 1, z_2 = 1$ in the original plumbing graph is replaced by a new sphere and two new edges shown in pink.

After determining the coefficients $\mathbb{V}$ and $\mathbf{V}$, one can use the L_1 conformal Ward identities

$$\begin{aligned} \hat{\rho}_f^{(\eta)}(L_1 v_{1,n_1}, v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3}) &= \hat{\rho}_f^{(\eta)}(v_{1,n_1}, L_{-1} v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3}) + \hat{\rho}_f^{(\eta)}(v_{1,n_1}, v_{2,n_2}^{\alpha_2}, L_{-1} v_{3,n_3}^{\alpha_3}), \\ \hat{\rho}_f^{(\eta)}(v_{1,n_1}, L_1 v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3}) &= \hat{\rho}_f^{(\eta)}(L_{-1} v_{1,n_1}, v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3}) - \hat{\rho}_f^{(\eta)}(v_{1,n_1}, L_{-1} v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3}) \\ &\quad - 2\hat{\rho}_f^{(\eta)}(v_{1,n_1}, L_0 v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3}) - \hat{\rho}_f^{(\eta)}(v_{1,n_1}, v_{2,n_2}^{\alpha_2}, L_1 v_{3,n_3}^{\alpha_3}), \end{aligned} \quad (28)$$

which become recursion relations for the branching coefficients $\mathbb{B}(n_1, n_2, n_3)$. Given the boundary values $\mathbb{B}(n_1, n_2, n_3)$ with $-\frac{3}{4} \leq n_i \leq \frac{3}{4}$, which can be computed analytically using the free field realization, one can solve for $\mathbb{B}(n_1, n_2, n_3)$ recursively.

Once one knows the branching coefficients, one can use the c -recursion relation to compute the Virasoro conformal blocks [8], which computes $\widehat{\mathbf{F}}_{\hat{\Gamma}}^{\{f,\eta,a\}}$ as a q -series. The free fermion block $\mathbf{F}_{\hat{\Gamma}}^F$ can be computed from its definition efficiently due to the small size of the space of fermion descendants. The superconformal block $\mathbf{F}_{\hat{\Gamma}}^{\{f,\eta,a\}}$, as a series in $\{\eta_e, q_e\}$, can be extracted coefficient-by-coefficient from (21).

5 Blocks with odd Ramond loops

We now turn to the case where the block contains odd Ramond loops. In this case, we define a new, non-vanishing enlarged block $\widehat{\mathbf{F}}_f^{(f,\eta,a)}[\Theta\psi]$ by the following: For each odd Ramond loop C , we split one of the edges e in C . Suppose e connects two Riemann spheres with complex

coordinates z, z' , and the punctures e connects to are located at $z = z_0, z' = z'_0$. We replace e by introducing a new Riemann sphere w , and glue the two punctures $w = 0, w = \infty$ with the two original spheres z_1, z_2 by

$$(z - z_0)w = q_{e_L}, \quad (z' - z'_0)w^{-1} = q_{e_R}. \quad (29)$$

The two new edges e_L, e_R formed are both Ramond. We repeat this splitting procedure for each odd Ramond loop in $\hat{\Gamma}$, and we denote the set of all split edges by E_{split} . This procedure defines a new plumbing channel Γ_{split} , whose plumbing parameters are given by $\{q_e\}_{e \notin E_{\text{split}}}$ and $\{q_{e_L}, q_{e_R}\}_{e \in E_{\text{split}}}$. The original parameter q_e for a split edge can be related to q_{e_L}, q_{e_R} by $q_e = q_{e_L} q_{e_R}$. Similarly, the spin sign η_e can be recovered by $\eta_e = \eta_{e_L} \eta_{e_R}$. An example of this procedure where one splits the second edge of the genus-2 theta channel is illustrated in figure 2.

On each new sphere w , we insert the operator $\Theta\psi$ at $w = 1$, where the operator Θ acts on the fermion ground states as

$$\Theta u^b = (-1)^{b+1} u^{1-b}, \quad (30)$$

and it anticommutes with ψ_r, G_r while commuting with L_n . The new enlarged block $\hat{\mathbf{F}}_f^{(f, \eta, a)}[\Theta\psi]$ is defined to be

$$\begin{aligned} \hat{\mathbf{F}}_{\hat{\Gamma}}^{\{f, \eta, a\}}[\Theta\psi] = & \sum_{\{A, \alpha, \mathbf{A}, \mathbf{a}\}} (-1)^{K_{\Gamma_{\text{split}}}(\epsilon + \epsilon')} \prod_{e \in E_{\text{NS}} \cup (E_R \setminus E_{\text{split}})} q_e^{|A_{e_1}| + |A_{e_1}|} \eta_e^{\epsilon_e + \epsilon'_e} (-1)^{\epsilon_e \epsilon'_e} \\ & \times \prod_{e \in E_{\text{NS}}} \mathbb{G}_{\mathbb{F}}^{A_{e_1} A_{e_2}} \mathbb{G}_{h_e}^{A_{e_1} A_{e_2}} \prod_{e \in E_R \setminus E_{\text{split}}} \mathbb{G}_{\mathbb{F}}^{A_{e_1} \mathbf{a}_{e_1}, A_{e_2} \mathbf{a}_{e_2}} \mathbb{G}_{\beta_e}^{A_{e_1} \alpha_{e_1}, A_{e_2} \alpha_{e_2}} \\ & \times \prod_{v \in V_{\text{NS}}} \hat{\rho}_{av}(\Psi_{-A_{v_1}} \mathbf{1} \otimes \mathbb{L}_{-A_{v_1}} \phi_{v_1}, \Psi_{-A_{v_2}} \mathbf{1} \otimes \mathbb{L}_{-A_{v_2}} \phi_{v_2}, \Psi_{-A_{v_3}} \mathbf{1} \otimes \mathbb{L}_{-A_{v_3}} \phi_{v_3}) \\ & \times \prod_{v \in V_R} \hat{\rho}_{fv}^{(\eta_v)}(\Psi_{-A_{v_1}} \mathbf{1} \otimes \mathbb{L}_{-A_{v_1}} \phi_{v_1}, \Psi_{-A_{v_2}} u^{\mathbf{a}_{v_2}} \otimes \mathbb{L}_{-A_{v_2}} w_{v_2}^{\alpha_{v_2}}, \Psi_{-A_{v_3}} u^{\mathbf{a}_{v_3}} \otimes \mathbb{L}_{-A_{v_3}} w_{v_3}^{\alpha_{v_3}}) \\ & \times \prod_{e \in E_{\text{split}}} q_{e_L}^{|A_{e_1}| + |A_{e_1}|} q_{e_R}^{|A_{e_2}| + |A_{e_2}|} (\eta_{e_L} \eta_{e_R})^{\epsilon_e + \epsilon'_e} \\ & \times \mathbb{G}_{\mathbb{F}}^{A_{e_1} \mathbf{a}_{e_1}, A_{e_L} \mathbf{a}_{e_L}} \mathbb{G}_{\beta_e}^{A_{e_1} \alpha_{e_1}, A_{e_L} \alpha_{e_L}} \mathbb{G}_{\mathbb{F}}^{A_{e_R} \mathbf{a}_{e_R}, A_{e_2} \mathbf{a}_{e_2}} \mathbb{G}_{\beta_e}^{A_{e_R} \alpha_{e_R}, A_{e_2} \alpha_{e_2}} \\ & \times \langle \langle \Psi_{-A_{e_2}} u^{\mathbf{a}_{e_2}} \otimes \mathbb{L}_{-A_{e_R}} w_{e_R}^{\alpha_{e_R}} | \Theta\psi(1) | \Psi_{-A_{e_1}} u^{\mathbf{a}_{e_1}} \otimes \mathbb{L}_{-A_{e_L}} w_{e_L}^{\alpha_{e_L}} \rangle \rangle, \end{aligned} \quad (31)$$

where the sum is over all displayed labels, subject to (7) and (18), except that the fermion labels A_{e_1}, A_{e_2} and $\mathbf{a}_{e_1}, \mathbf{a}_{e_2}$ on a split edge may differ due to the $\psi(1)$ insertion. Their combined parities are constrained to be the same, namely

$$(-1)^{A_{e_1} + \mathbf{a}_{e_1}} = (-1)^{A_{e_2} + \mathbf{a}_{e_2}}, \quad e \in E_{\text{split}}. \quad (32)$$

This enlarged block is non-vanishing, and can be expressed as a convolution of the ordinary superconformal block and the free fermion block $\mathbf{F}_{\hat{\Gamma}}^{\mathbb{F}}[\Theta\psi]$ with the same insertion $\Theta\psi(1)$ on the

inserted spheres of each split edge. Namely,

$$\widehat{\mathbf{F}}_{\hat{\Gamma}}^{\{f,\eta,a\}}[\Theta\psi] = \mathbf{F}_{\hat{\Gamma}}^{\{f,\eta,a\}} \star_{\hat{\Gamma}} \mathbf{F}_{\hat{\Gamma}}^{\mathbf{F}}[\Theta\psi], \quad (33)$$

where the convolution is defined from the unsplitted channel $\hat{\Gamma}$, and one restrict to the terms in $\widehat{\mathbf{F}}_{\hat{\Gamma}}^{\{f,\eta,a\}}[\Theta\psi]$, $\mathbf{F}_{\hat{\Gamma}}^{\mathbf{F}}[\Theta\psi]$ that have same powers of q_{e_L} , q_{e_R} and same powers of η_{e_L} , η_{e_R} for each split edge e , and replace $q_{e_L}^m q_{e_R}^m$ by q_e^m , $\eta_{e_L}^\epsilon \eta_{e_R}^\epsilon$ by η_e^ϵ in these terms before performing the convolution.

On the other hand, since the operator ψ is proportional to $v_{1/2}(P)$ with Liouville momentum $P = \frac{Q}{2}$, and the operator Θ commutes with the two copies of Virasoro algebras, one can decompose $\widehat{\mathbf{F}}_f^{(\eta,\eta')}[\Theta\psi]$ into Virasoro blocks as

$$\begin{aligned} \widehat{\mathbf{F}}_{\hat{\Gamma}}^{\{f,\eta,a\}}[\Theta\psi] &= \sum_{\{n,\alpha\}} (-1)^{K_{\Gamma_{\text{split}}}(\epsilon)} \prod_{e \in E_{\text{NS}}} q_e^{2n_e^2} \eta_e^{2n_e} \prod_{e \in E_{\text{R}} \setminus E_{\text{split}}} q_e^{2n_e^2 - \frac{1}{8}} \eta_e^{\alpha_e} \\ &\times \prod_{e \in E_{\text{split}}} q_{e_L}^{2n_{e_L}^2 - \frac{1}{8}} q_{e_R}^{2n_{e_R}^2 - \frac{1}{8}} \eta_e^{\alpha_e} \mathbb{S}(n_{e_R}, n_{e_L}, \alpha_e) \\ &\times \prod_{i=1}^2 F_{\Gamma'}(h_{e,n_e}^{(i)}, h_{\psi}^{(i)}, c^{(i)}) \prod_{v \in V_{\text{NS}}} \mathbf{B}_{a_v}(n_{v_1}, n_{v_2}, n_{v_3}) \\ &\times \prod_{v \in V_{\text{R}}} \mathbb{B}_{f_v}^{(\eta_v)}(n_{v_1}, n_{v_2}, n_{v_3}, \alpha_{v_2}, \alpha_{v_3}), \end{aligned} \quad (34)$$

where $F_{\Gamma'}(h_e^{(i)}, h_{\psi}^{(i)}, c^{(i)})$ stands for Virasoro conformal blocks in the new channel Γ' with external conformal weights

$$h_{\psi}^{(1)} = -\frac{1+2b^2}{2(1-b^2)}, \quad h_{\psi}^{(2)} = \frac{b^2+2}{2(1-b^2)} \quad (35)$$

on the punctures with $\Theta\psi$ insertions. The splitting coefficient $\mathbb{S}(n_R, n_L, \alpha)$ is defined as

$$\mathbb{S}(n_R, n_L, \alpha) = \frac{\langle \langle v_{n_R}^\alpha | \Theta\psi(1) | v_{n_L}^\alpha \rangle \rangle}{\|v_{n_L}^\alpha\| \|v_{n_R}^\alpha\|}, \quad (36)$$

which is nonvanishing only for $n_R = n_L \pm \frac{1}{2}$.

Once the coefficients $\mathbb{V}_{n,\alpha}^{1,2}$, $\mathbb{V}_{\pm 1,n,\alpha}^{AB}$ in (27) are known, the splitting coefficients can be solved for using the L_1 conformal Ward identity

$$\langle \langle L_1 v_{n_R}^\alpha | \Theta\psi(1) | v_{n_L}^\alpha \rangle \rangle = \langle \langle v_{n_R}^\alpha | \Theta\psi(1) | L_{-1} v_{n_L}^\alpha \rangle \rangle, \quad (37)$$

which becomes a recursion relation for the splitting coefficients after plugging in (27). After solving for the splitting coefficients and the branching coefficients, one can compute $\widehat{\mathbf{F}}_{\hat{\Gamma}}^{\{f,\eta,a\}}[\Theta\psi]$ using this decomposition, and the physical superconformal block $\mathbf{F}_{\hat{\Gamma}}^{\{f,\eta,a\}}$ can be extracted from (33). It can be tested numerically that the superconformal block computed this way only depends on the product $q_e = q_{e_L} q_{e_R}$, but not the individual parameters q_{e_L} , q_{e_R} of the split edges.

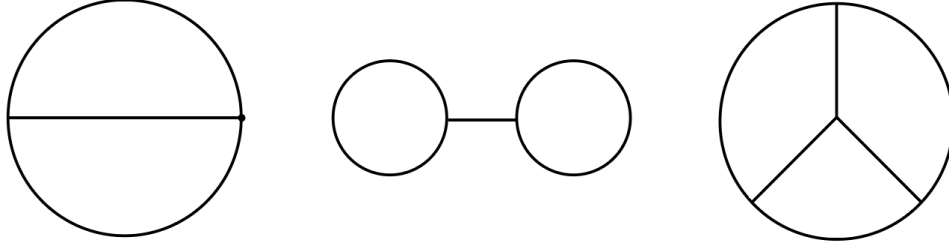


Figure 3: From left to right: The genus-2 theta channel, the genus-2 glasses channel, and the genus-3 tetrahedron channel.

6 Numerical Tests

Using the algorithm presented in Sections 4 and 5, one can compute arbitrary superconformal blocks as a series in the plumbing parameters. We have tested our algorithm in the genus-2 theta and glasses channels, as well as the genus-3 tetrahedron channel. The plumbing graphs of these channels are shown in figure 3.

On a laptop and using 40-digit working precision, computing the genus-2 theta channel block with NS-R-R edges takes 7 minutes up to level $n_1, n_2, n_3 \leq 10$ in the case without an odd Ramond loop, and takes 27 minutes for the case with an odd Ramond loop. A direct computation using the definition (6) is estimated to take more than 6 hours for the same block on the same device with the same precision.

For all the spin structures in the two genus-2 channels, the low-level coefficients of $q_1^{n_1} q_2^{n_2} q_3^{n_3}$ in $\mathbf{F}_{\hat{\Gamma}}^{\{f, \eta, a\}}$ obtained from our algorithm agree with direct computations from the definition (6) up to level $n_1, n_2, n_3 \leq 5$ for all combinations of f_v, η_v, a_v . In the theta channel and the tetrahedron channel with all NS internal edges, the superconformal block computed from our algorithm agrees with the result from c -recursion [9] up to level $n_1, n_2, n_3 \leq 10$. The same test has been performed for the genus-3 tetrahedron channel, where the edges around one face are Ramond while the other edges are NS. For this channel, the block $\mathbf{F}_{\hat{\Gamma}}^{\{f, \eta, a\}}$ agrees with its definition up to total level $n_1 + n_2 + n_3 \leq 5$, and agrees with the result of c -recursion up to total level $n_1 + n_2 + n_3 \leq 10$.

Moreover, there are plumbing data $\{q_e, \eta_e\}$ in the all-NS theta channel and the NS-R-R theta channel that describe the same spin Riemann surface, and we have checked that the ratio

$$\frac{Z_{\text{super-Liouville}}}{(Z_{\text{free boson+fermion}})^{1+2Q^2}} \quad (38)$$

of super-Liouville and free boson + fermion partition functions agrees in the two plumbing channels when evaluated using our superconformal blocks. The details of this check are ex-

plained in Appendix C. The codes that implement our algorithm and perform the mentioned cross checks are available in a GitHub repository ³.

7 Discussion

In this work, we develop the algorithm for numerically computing superconformal blocks at arbitrary genus, in arbitrary plumbing channels, for arbitrary spin structures. The double Virasoro subalgebra allows us to decompose the enlarged $\text{SCA}_\nu \oplus \text{F}_\nu$ block into Virasoro conformal blocks, with the coefficients given by branching coefficients, which are known on an (NS, NS, NS) sphere and can be solved numerically using a recursion relation on an (NS, R, R) sphere.

Our algorithm provides new tools for higher-loop superstring amplitudes. For 2-dimensional Type 0B and Type 0A string theory, such amplitudes can be used to further test the duality with matrix quantum mechanics [17–20], as well as the T-duality between the two theories [21]. They also permit direct worldsheet tests of matrix-integral predictions for recently proposed super Virasoro minimal strings [22–26]. One particularly interesting direction is the 2-dimensional $\text{SO}(23)$ heterotic string [27], where the matrix quantum mechanical dual remains unknown, and the higher-loop or higher-point amplitudes can be used to find it. In these applications of string amplitudes, our algorithm can be used to compute the integrand on moduli space. The integration can be performed using a similar method to [28]: For genus $g \leq 3$, one samples the moduli space parametrized by the period matrix and converts the period matrix into plumbing parameters using a similar algorithm to [29].

Beyond string amplitudes, modular covariance of higher-genus partition functions can be used to constrain the spectrum and the structure constants of $\mathcal{N} = 1$ SCFTs, extending the analysis of [30]. The superconformal blocks themselves are conjectured to be the wavefunctions of 3D supergravity in AdS_3 [26], and the numerical values of the blocks obtained from our algorithm can be used to compute gravitational path integrals in this theory [31].

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³Link to github

A Conventions

In this appendix, we summarize the choices of convention used in this paper.

The superconformal algebra The $\mathcal{N} = 1$ superconformal algebra is given by

$$\begin{aligned} [L_m, L_n] &= (m - n)L_{m+n} + \frac{c}{12} (m^3 - m) \delta_{m+n,0}, \\ [L_m, G_r] &= \left(\frac{m}{2} - r\right) G_{m+r}, \\ \{G_r, G_s\} &= 2L_{r+s} + \frac{c}{3} \left(r^2 - \frac{1}{4}\right) \delta_{r+s,0}. \end{aligned} \quad (39)$$

BPZ conjugation The BPZ conjugation are defined using the inversion map $I : z \mapsto w = 1/z$, and we choose the spin lift such that

$$\left(\frac{dw}{dz}\right)^{\frac{1}{2}} = -\frac{i}{z}, \quad (40)$$

which is consistent with [Sen]. On each plumbed edge with plumbing map $zw = q$, we choose the spin lift such that

$$\left(\frac{dw}{dz}\right)^{\frac{1}{2}} = -\frac{i\sqrt{q}}{z}, \quad (41)$$

with the branch of $\sqrt{q}$ chosen such that $\sqrt{q}$ for positive real q gives the positive square root. Under BPZ conjugation, the superconformal generators and the auxiliary fermion transforms as

$$L_n^\dagger = L_{-n}, \quad G_r^\dagger = iG_{-r}, \quad \psi_r^\dagger = -i\psi_{-r}. \quad (42)$$

We also adopt the convention that the BPZ conjugation acts on product as

$$(XY)^\dagger = (-1)^{\epsilon_X \epsilon_Y} Y^\dagger X^\dagger, \quad (43)$$

where ϵ_X, ϵ_Y are the Grassmann parities of X, Y . For tensor product operators, the BPZ inner product is given by

$$\langle\langle A \otimes X | B \otimes Y \rangle\rangle = (-1)^{\epsilon_A \epsilon_X} \langle\langle A | B \rangle\rangle \langle\langle X | Y \rangle\rangle. \quad (44)$$

Ward identities The L_n Ward identity is given by

$$\begin{aligned} &\sum_{j \geq 0} (-1)^j \binom{n}{j} \rho(L_{j-m-n+1} x_1, x_2, x_3) \\ &= \sum_{j \geq 0} \binom{m}{j} \rho(x_1, L_{n+j-1} x_2, x_3) + (-1)^n \sum_{j \geq 0} (-1)^j \binom{n}{j} \rho(x_1, x_2, L_{m+j-1} x_3). \end{aligned} \quad (45)$$

Here, ρ is either ρ_a or $\rho_f^{(\eta)}$. The G_r and ψ_r Ward identities require one to choose a branch for the square root $\sqrt{z(z-1)}$, which we choose to be the following

$$\sqrt{z(z-1)} = \begin{cases} z\sqrt{1-z^{-1}}, & z \sim \infty, \\ \sqrt{z-1}\sqrt{1+(z-1)}, & z \sim 1, \\ i\sqrt{z}\sqrt{1-z}, & z \sim 0. \end{cases} \quad (46)$$

With this choice of branch, the G_r and ψ_r Ward identities are given by

$$\begin{aligned} & \sum_{j \geq 0} (-1)^j \binom{n}{j} \rho_a(G_{j-m-n+\frac{1}{2}} x_1, x_2, x_3) \\ &= i(-1)^{\epsilon_1} \sum_{j \geq 0} \binom{m}{j} \rho_a(x_1, G_{n+j-\frac{1}{2}} x_2, x_3) + i(-1)^{\epsilon_1+\epsilon_2+n} \sum_{j \geq 0} (-1)^j \binom{n}{j} \rho_a(x_1, x_2, G_{m+j-\frac{1}{2}} x_3), \end{aligned} \quad (47)$$

$$\begin{aligned} & \sum_{j \geq 0} (-1)^j \binom{n}{j} \rho_F(\psi_{j-m-n-\frac{1}{2}} x_1, x_2, x_3) \\ &= -i(-1)^{\epsilon'_1} \sum_{j \geq 0} \binom{m}{j} \rho_F(x_1, \psi_{n+j+\frac{1}{2}} x_2, x_3) - i(-1)^{\epsilon'_1+\epsilon'_2+n} \sum_{j \geq 0} (-1)^j \binom{n}{j} \rho_F(x_1, x_2, \psi_{m+j+\frac{1}{2}} x_3). \end{aligned} \quad (48)$$

$$\begin{aligned} & \sum_{j \geq 0} (-1)^j \binom{n+\frac{1}{2}}{j} \rho_f^{(\eta)}(G_{j-m-n-\frac{1}{2}} x_1, x_2, x_3) \\ &= i(-1)^{\epsilon_1} \sum_{j \geq 0} \binom{m+\frac{1}{2}}{j} \rho_f^{(\eta)}(x_1, G_{n+j} x_2, x_3) - (-1)^{\epsilon_1+f+\epsilon_2+n} \sum_{j \geq 0} (-1)^j \binom{n+\frac{1}{2}}{j} \rho_f^{(\eta)}(x_1, x_2, G_{m+j} x_3), \end{aligned} \quad (49)$$

$$\begin{aligned} & \sum_{j \geq 0} (-1)^j \binom{n-\frac{1}{2}}{j} \rho_F(\psi_{j-m-n+\frac{1}{2}} x_1, x_2, x_3) \\ &= -i(-1)^{\epsilon'_1} \sum_{j \geq 0} \binom{m-\frac{1}{2}}{j} \rho_F(x_1, \psi_{n+j} x_2, x_3) - (-1)^{\epsilon'_1+\epsilon'_2+n} \sum_{j \geq 0} (-1)^j \binom{n-\frac{1}{2}}{j} \rho_F(x_1, x_2, \psi_{m+j} x_3). \end{aligned} \quad (50)$$

B Free field representation of $\text{Vir} \oplus \text{Vir}$ primaries

The $\text{Vir} \oplus \text{Vir}$ primaries can be directly constructed using a free field realization of the $\text{SCA}_\nu \oplus \text{F}_\nu$ algebra [16]. Using free boson operators $\{c_n | n \neq 0\}$, free fermion operators $\{\eta_r | r \in \mathbb{Z} + \nu\}$ and the central element $\hat{P}$ that satisfy the algebraic relations

$$[c_n, c_m] = n\delta_{m+n,0}, \quad \{\eta_r, \eta_s\} = \delta_{r+s,0}, \quad (51)$$

we can realize the generators of the superconformal algebra SCA_ν as

$$\begin{aligned} L_{n \neq 0} &= \frac{1}{2} \sum_{k \neq 0, n} c_k c_{n-k} + \frac{1}{2} \sum_r r \eta_{n-r} \eta_r + \frac{i}{2} (Qn - 2\hat{P}) c_n, \\ L_0 &= \sum_{k > 0} c_{-k} c_k + \sum_{r > 0} r \eta_{-r} \eta_r + \frac{1}{2} \left(\frac{Q^2}{4} - \hat{P}^2 \right) + \frac{1-2\nu}{16}, \\ G_r &= \sum_{n \neq 0} c_n \eta_{r-n} + i(Qr - \hat{P}) \eta_r. \end{aligned} \quad (52)$$

Each free boson mode c_n and free fermion mode η can be expressed directly in terms of G_{-r} , L_{-n} and $\hat{P}$.

It can be proved that $\mathcal{M}_{\text{NS}}(h)$ is isomorphic to the Fock space generated by acting with $c_{n < 0}, \psi_{r < 0}, \eta_{r < 0}$ on a ground state $\phi(P)$ that satisfies $\hat{P}\phi(P) = P\phi(P)$ ⁴, and the $\text{Vir} \oplus \text{Vir}$ primaries in $\mathcal{M}_{\text{NS}}(h)$ can be constructed using the free field operators $\chi_r(P) = \psi_r - i\eta_r(P)$ as

$$v_n(P) = \chi_{-\frac{4n-1}{2}}(P) \chi_{-\frac{4n-3}{2}}(P) \cdots \chi_{-\frac{1}{2}}(P) \phi(P) \quad (53)$$

where $\eta_r(P)$ is the solution of η_r in terms of $L_n, G_r, \hat{P}$ and with $\hat{P}$ replaced by P . v_n with negative n is defined by the reflection formula

$$v_n(P) = v_{-n}(-P). \quad (54)$$

Similarly in the R-sector, the $\text{Vir} \oplus \text{Vir}$ primaries in $\mathcal{M}_{\text{R}}(\beta)$ can be constructed by

$$\begin{aligned} v_n^{2n+\frac{1}{2} \bmod 2}(P) &= \chi_0(P) \chi_{-1}(P) \cdots \chi_{-(2n-\frac{1}{2})}(P) (u^0 \otimes w_\beta^+), \\ v_n^{2n-\frac{1}{2} \bmod 2}(P) &= \chi_0(P) \chi_{-1}(P) \cdots \chi_{-(2n-\frac{1}{2})}(P) \chi_0(-P) (u^0 \otimes w_\beta^+), \end{aligned} \quad (55)$$

for $n > 0$, and by the reflection formula

$$(v_n^\alpha(P))_+ = (v_{-n}^\alpha(-P))_+, \quad (v_n^\alpha(P))_- = -(v_{-n}^\alpha(-P))_-, \quad (56)$$

for $n < 0$. The subscripts $\pm$ denote the components ending in w_β^+ and w_β^- , respectively.

The norms of these $\text{Vir} \oplus \text{Vir}$ primaries are computed in [16]. We define the function $\ell(x, m)$ to be

$$\ell(x, m) = 2^{(1-(-1)^m)/16} \prod_{\substack{r, s \geq 0, \ r+s < m \\ r+s \equiv m \pmod{2}}} (x + rb + s/b), \quad m \geq 0, \quad (57)$$

⁴We take the branch of P as follows: For h smaller than $Q^2/8$, P is taken to be the positive root of $\frac{1}{4}Q^2 - 2h$. For h greater than $Q^2/8$, we take $P = ip$ such that $p > 0$. In the R-sector, P is taken to be the positive root of $2\beta^2$ for $\beta^2 > 0$ and the root with positive imaginary part for $\beta^2 < 0$

and $\ell(x, 0) = 1$. For $m < 0$, we define their $\ell(x, m)$ using the reflection formula

$$\ell(x, -m) = \begin{cases} \ell(Q - x, m), & m > 0 \text{ odd}, \\ (-1)^{m/2} \ell(Q - x, m), & m > 0 \text{ even}. \end{cases} \quad (58)$$

The BPZ norms of the $\text{Vir} \oplus \text{Vir}$ primaries with $n > 0$ are given by

$$\begin{aligned} \|v_n(P)\|^2 &= i^{2n \bmod 2} (-1)^{2n} 2^{2n} \frac{\ell(2P, 4n)}{\ell(Q + 2P, 4n)}, \\ \|v_n^0(P)\|^2 &= 2^{2[M/2]+1} \frac{\ell(2P, 4n)}{\ell(Q + 2P, 4n)}, \\ \|v_n^1(P)\|^2 &= -i 2^{2[M/2]} \frac{\ell(2P, 4n)}{\ell(Q + 2P, 4n)}, \end{aligned} \quad (59)$$

where $M = 2n - \frac{1}{2}$. For the norms of the $\text{Vir} \oplus \text{Vir}$ primaries with $n < 0$, one replace P with $-P$, n with $-n$ and use the formula above.

The (NS, NS, NS) branching coefficient is also fixed in [16] using the free field representation. In our convention, their result reads

$$\begin{aligned} \mathbf{B}_a(n_1, n_2, n_3) &= \frac{(-1)^{2n_3} (-i)^a e^{\frac{i\pi}{4}(\epsilon_1 - \epsilon_2 - \epsilon_3)}}{\prod_{i=1}^3 \sqrt{\ell(2P_i, 4n_i) \ell(Q + 2P_i, 4n_i)}} \\ &\times \ell\left(\frac{Q}{2} + P_1 + P_2 + P_3, 2(n_1 + n_2 + n_3)\right) \ell\left(\frac{Q}{2} - P_1 + P_2 + P_3, 2(-n_1 + n_2 + n_3)\right) \\ &\times \ell\left(\frac{Q}{2} + P_1 + P_2 - P_3, 2(n_1 + n_2 - n_3)\right) \ell\left(\frac{Q}{2} - P_1 + P_2 - P_3, 2(-n_1 + n_2 - n_3)\right), \end{aligned} \quad (60)$$

where $\epsilon_i = 2n_i \bmod 2$. In the code, we used this formula directly for the (NS, NS, NS) branching coefficient, while the (NS, R, R) ones are obtained from the L_1 Ward identities.

C The cross-channel test

In this appendix, we give details to the cross-channel check mentioned in Section 6. We consider the $\mathcal{N} = 1$ super-Liouville theory with $b = \frac{7}{5}$, $Q = b + b^{-1}$, and the central charge $c = \frac{3}{2}(1 + 2Q^2)$.

The spectrum of this theory is given by

$$h_{\text{NS}}(p) = \frac{Q^2}{8} + \frac{p^2}{2}, \quad h_{\text{R}}(p) = \frac{Q^2}{8} + \frac{p^2}{2} + \frac{1}{16}, \quad (61)$$

with p ranging from 0 to ∞ . The (NS, NS, NS) structure constants $C_a(p_1, p_2, p_3)$ are given by

$$\begin{aligned} C_0(p_1, p_2, p_3) &= \frac{\Upsilon'_{\text{NS}}(0)}{b^3} \frac{\prod_{j=1}^3 \sqrt{\Upsilon_{\text{NS}}(2ip_j) \Upsilon_{\text{NS}}(-2ip_j)}}{\Upsilon_{\text{NS}}(Q/2 + i \sum_j p_j) \prod_{j=1}^3 \Upsilon_{\text{NS}}(Q/2 + i \sum_k p_k - 2ip_j)}, \\ C_1(p_1, p_2, p_3) &= \frac{2i \Upsilon'_{\text{NS}}(0)}{b^3} \frac{\prod_{j=1}^3 \sqrt{\Upsilon_{\text{NS}}(2ip_j) \Upsilon_{\text{NS}}(-2ip_j)}}{\Upsilon_{\text{R}}(Q/2 + i \sum_j p_j) \prod_{j=1}^3 \Upsilon_{\text{R}}(Q/2 + i \sum_k p_k - 2ip_j)}, \end{aligned} \quad (62)$$

where Υ is the Barnes double gamma function, and

$$\Upsilon_{\text{NS}}(x) = \Upsilon_b(x/2)\Upsilon_b((x+Q)/2), \quad \Upsilon_{\text{R}}(x) = \Upsilon_b((x+b)/2)\Upsilon_b((x+b^{-1})/2). \quad (63)$$

The (NS, R, R) structure constants $C_{0,\eta}$ are given by

$$C_{0,+}(p_1, p_2, p_3) = \frac{\Upsilon'_{\text{NS}}(0)}{b^3} \frac{\sqrt{\prod_{\sigma=\pm 1} \Upsilon_{\text{NS}}(2i\sigma p_1)\Upsilon_{\text{R}}(2i\sigma p_2)\Upsilon_{\text{R}}(2i\sigma p_3)}}{\prod_{\sigma=\pm 1} \Upsilon_{\text{R}}(Q/2 + i(p_2 + p_3 + \sigma p_1))\Upsilon_{\text{NS}}(Q/2 + i(p_1 + \sigma(p_2 - p_3)))},$$

$$C_{0,-}(p_1, p_2, p_3) = \frac{\Upsilon'_{\text{NS}}(0)}{b^3} \frac{\sqrt{\prod_{\sigma=\pm 1} \Upsilon_{\text{NS}}(2i\sigma p_1)\Upsilon_{\text{R}}(2i\sigma p_2)\Upsilon_{\text{R}}(2i\sigma p_3)}}{\prod_{\sigma=\pm 1} \Upsilon_{\text{NS}}(Q/2 + i(p_2 + p_3 + \sigma p_1))\Upsilon_{\text{R}}(Q/2 + i(p_1 + \sigma(p_2 - p_3)))}.$$

The other two coefficients obey $C_{1,\eta} = C_{0,\eta}$.

We will focus on the genus-2 partition function of this theory in the genus-2 theta channel. In the theta channel with NS-NS-NS edges, the a label in both 3-point functions ρ_a are forced to be the same by the Gram matrices. We can then denote the all-NS theta channel superconformal block as $\mathbf{F}^{(a)}(h_{1,2,3}; q_{1,2,3}, \eta_{1,2,3})$. For the theta channel with NS-R-R edges, the two f labels of $\rho_f^{(\eta)}$ are forced to be the same, but the two η labels can be different. We will denote the NS-R-R theta channel block as $\mathbf{F}^{(f,\eta,\eta')}(h_1, \beta_{2,3}; q_{1,2,3}, \eta_{1,2,3})$, where η, η' are the η -label of the 3-point functions on the z_1 and the z_2 sphere. There is an odd Ramond loop in this channel if $\eta\eta' = -1$. Using these blocks, the super-Liouville partition function can be written as

The pairs of plumbing parameters in the NS-NS-NS and the NS-R-R

References

- [1] A. B. Zamolodchikov. “CONFORMAL SYMMETRY IN TWO-DIMENSIONS: AN EXPLICIT RECURRENCE FORMULA FOR THE CONFORMAL PARTIAL WAVE AMPLITUDE”. In: *Commun. Math. Phys.* 96 (1984), pp. 419–422.
- [2] Al. B. Zamolodchikov. “Conformal symmetry in two-dimensional space: Recursion representation of conformal block”. In: *Theor. Math. Phys.* 73.1 (1987), pp. 1088–1093.
- [3] Leszek Hadasz, Zbigniew Jaskolski, and Paulina Suchanek. “Recursion representation of the Neveu-Schwarz superconformal block”. In: *JHEP* 03 (2007), p. 032. arXiv: [hep-th/0611266](#).
- [4] Leszek Hadasz, Zbigniew Jaskolski, and Paulina Suchanek. “Elliptic recurrence representation of the $N = 1$ Neveu-Schwarz blocks”. In: *Nucl. Phys. B* 798 (2008), pp. 363–378. arXiv: [0711.1619 \[hep-th\]](#).
- [5] V. A. Belavin. “ $N=1$ supersymmetric conformal block recursion relations”. In: *Theor. Math. Phys.* 152 (2007), pp. 1275–1285. arXiv: [hep-th/0611295](#).

- [6] Leszek Hadasz, Zbigniew Jaskolski, and Paulina Suchanek. “Elliptic recurrence representation of the $N=1$ superconformal blocks in the Ramond sector”. In: *JHEP* 11 (2008), p. 060. arXiv: 0810.1203 [hep-th].
- [7] Paulina Suchanek. “Elliptic recursion for 4-point superconformal blocks and bootstrap in $N=1$ SLFT”. In: *JHEP* 02 (2011), p. 090. arXiv: 1012.2974 [hep-th].
- [8] Minjae Cho, Scott Collier, and Xi Yin. “Recursive Representations of Arbitrary Virasoro Conformal Blocks”. In: *JHEP* 04 (2019), p. 018. arXiv: 1703.09805 [hep-th].
- [9] Vladimir Belavin and Roman Geiko. “c-Recursion for multi-point superconformal blocks. NS sector”. In: *JHEP* 08 (2018), p. 112. arXiv: 1806.09563 [hep-th].
- [10] Leszek Hadasz, Zbigniew Jaskolski, and Paulina Suchanek. “Recurrence relations for toric $N=1$ superconformal blocks”. In: *JHEP* 09 (2012), p. 122. arXiv: 1207.5740 [hep-th].
- [11] M. A. Bershtein and A. I. Shchekkin. “Bilinear equations on Painlevé τ functions from CFT”. In: *Commun. Math. Phys.* 339.3 (2015), pp. 1021–1061. arXiv: 1406.3008 [math-ph].
- [12] M. A. Bershtein and A. I. Shchekkin. “Backlund transformation of Painlevé III(D_8) tau function”. In: *J. Phys. A* 50.11 (2017), p. 115205. arXiv: 1608.02568 [math-ph].
- [13] Luis F. Alday, Davide Gaiotto, and Yuji Tachikawa. “Liouville Correlation Functions from Four-dimensional Gauge Theories”. In: *Lett. Math. Phys.* 91 (2010), pp. 167–197. arXiv: 0906.3219 [hep-th].
- [14] A. A. Belavin, M. A. Bershtein, B. L. Feigin, A. V. Litvinov, and G. M. Tarnopolsky. “Instanton moduli spaces and bases in coset conformal field theory”. In: *Commun. Math. Phys.* 319 (2013), pp. 269–301. arXiv: 1111.2803 [hep-th].
- [15] Volker Schomerus and Paulina Suchanek. “Liouville’s Imaginary Shadow”. In: *JHEP* 12 (2012), p. 020. arXiv: 1210.1856 [hep-th].
- [16] Leszek Hadasz and Zbigniew Jaskólski. “Super-Liouville - Double Liouville correspondence”. In: *JHEP* 05 (2014), p. 124. arXiv: 1312.4520 [hep-th].
- [17] Tadashi Takayanagi and Nicolaos Toumbas. “A Matrix model dual of type 0B string theory in two-dimensions”. In: *JHEP* 07 (2003), p. 064. arXiv: hep-th/0307083.
- [18] Michael R. Douglas, Igor R. Klebanov, D. Kutasov, Juan Martin Maldacena, Emil John Martinec, and N. Seiberg. “A New hat for the $c=1$ matrix model”. In: *From Fields to Strings: Circumnavigating Theoretical Physics: A Conference in Tribute to Ian Kogan*. July 2003, pp. 1758–1827. arXiv: hep-th/0307195.
- [19] Bruno Balthazar, Victor A. Rodriguez, and Xi Yin. “The S-matrix of 2D type 0B string theory. Part I. Perturbation theory revisited”. In: *JHEP* 05 (2023), p. 234. arXiv: 2201.05621 [hep-th].

- [20] Bruno Balthazar, Victor A. Rodriguez, and Xi Yin. “The S-matrix of 2D type 0B string theory. Part II. D-instanton effects”. In: *JHEP* 05 (2023), p. 235. arXiv: 2204.01747 [hep-th].
- [21] Xi Yin. “Matrix models, integrable structures, and T duality of type 0 string theory”. In: *Nucl. Phys. B* 714 (2005), pp. 137–171. arXiv: hep-th/0312236.
- [22] Clifford V. Johnson. “Supersymmetric Virasoro minimal strings”. In: *Phys. Rev. D* 110.6 (2024), p. 066016. arXiv: 2401.08786 [hep-th].
- [23] Clifford Johnson V. “Further aspects of Supersymmetric Virasoro Minimal Strings”. In: *SciPost Phys.* 21 (2026), p. 040. arXiv: 2506.19000 [hep-th].
- [24] Clifford Johnson V. “Supersymmetric Virasoro Minimal Strings: Unified Perturbative and Non-Perturbative Treatment”. In: (Aug. 2026). arXiv: 2608.20529 [hep-th].
- [25] Mukund Rangamani and Jianming Zheng. “Towards the super Virasoro minimal string”. In: *JHEP* 09 (2025), p. 061. arXiv: 2505.08892 [hep-th].
- [26] Lorenz Eberhardt. “The Super Virasoro Minimal String from 3d Supergravity”. In: (Apr. 2026). arXiv: 2604.26038 [hep-th].
- [27] Joshua L. Davis, Finn Larsen, and Nathan Seiberg. “Heterotic strings in two dimensions and new stringy phase transitions”. In: *JHEP* 08 (2005), p. 035. arXiv: hep-th/0505081.
- [28] Sam Christian, Ben Mazel, Yuchen Wang, Xi Yin, and Yutai Zhang. “Upcoming work”. In: ().
- [29] Sam Christian, Ben Mazel, Yuchen Wang, Xi Yin, and Yutai Zhang. “Bosonic String Amplitudes From Ribbon Graphs”. In: (Sept. 2026). arXiv: 2609.17660 [hep-th].
- [30] Minjae Cho, Scott Collier, and Xi Yin. “Genus Two Modular Bootstrap”. In: *JHEP* 04 (2019), p. 022. arXiv: 1705.05865 [hep-th].
- [31] Arpan Bhattacharyya, Saptaswa Ghosh, Poulami Nandi, and Sounak Pal. “3D $\mathcal{N} = 1$ supergravity from Virasoro TQFT: gravitational partition function and Out-of-time-order correlator”. In: *JHEP* 02 (2025), p. 027. arXiv: 2408.01538 [hep-th].